

KITABU QUEST

S4

COMPUTER SCIENCE



Cover scene

students using computers and code safely with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

4

Title Page

KITABU QUEST S4 Computer Science

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S4 learners study Computer Science one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Practise safe computing, algorithms, data, systems thinking, and clear step-by-step explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Computer fundamental
2. Safe and ethical use of computer
3. Number system
4. Boolean algebra and logic gates
5. Control statements and one dimension array
6. Expression and Operator in C++ language
7. Control Statements in C++
8. Function in C++ language
9. Arrays in C++
10. HTML
11. Cascading style sheet

As you finish each topic, return to the success criteria and correct one answer before moving on.

Computer fundamental: Lesson Opener

Learning focus: Computer fundamental

Country link: a safe Rwanda observation, model, data table, or investigation for Computer fundamental.

By the end of this topic, you should be able to:

- explain characteristics and evolution of computers and detect the impact of computer in society

Inquiry questions:

- How can computer fundamental help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer fundamental: Learn

Computer fundamental teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Computer fundamental
- Computer
- fundamental
- concept
- observation
- evidence
- safety
- conclusion

Computer fundamental: Worked Example

Worked example for Computer fundamental: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Computer fundamental: Vocabulary And Study Skill

Use these words and study moves before you practise Computer fundamental. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Computer fundamental
- Computer
- fundamental
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Computer fundamental without a clear example, method, or evidence.

Computer fundamental: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Computer fundamental.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Computer fundamental: Practice And Correction

Practice for Computer fundamental:

Main outcome: explain characteristics and evolution of computers and detect the impact of computer in society.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Computer fundamental: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Computer fundamental.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer fundamental: Outcome Check

Outcome check for Computer fundamental: Explain characteristics and evolution of computers and detect the impact of computer in society.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Safe and ethical use of computer: Lesson Opener

Learning focus: Safe and ethical use of computer

Country link: a safe Rwanda observation, model, data table, or investigation for Safe and ethical use of computer.

By the end of this topic, you should be able to:

- integrate safety guidelines, ergonomics and ethical issues to have a good working environment

Inquiry questions:

- How can safe and ethical use of computer help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Safe and ethical use of computer: Learn

Safe and ethical use of computer teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Safe and ethical use of computer
- Safe
- ethical
- computer
- concept
- observation
- evidence
- safety

Safe and ethical use of computer: Worked Example

Worked example for Safe and ethical use of computer: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Safe and ethical use of computer: Vocabulary And Study Skill

Use these words and study moves before you practise Safe and ethical use of computer. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Safe and ethical use of computer
- Safe
- ethical
- computer
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Safe and ethical use of computer without a clear example, method, or evidence.

Safe and ethical use of computer: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Safe and ethical use of computer.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Safe and ethical use of computer: Practice And Correction

Practice for Safe and ethical use of computer:

Main outcome: integrate safety guidelines, ergonomics and ethical issues to have a good working environment.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Safe and ethical use of computer: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Safe and ethical use of computer.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Safe and ethical use of computer: Outcome Check

Outcome check for Safe and ethical use of computer: Integrate safety guidelines, ergonomics and ethical issues to have a good working environment.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Number system: Lesson Opener

Learning focus: Number system

Country link: a safe Rwanda observation, model, data table, or investigation for Number system.

By the end of this topic, you should be able to:

- compute numbers in different base systems and to do arithmetic operations based on binary numbers

Inquiry questions:

- How can number system help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Number system: Learn

Number system is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Number system
- Number
- system
- concept
- observation
- evidence
- safety
- conclusion

Number system: Worked Example

Investigation model for Number system: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Number system: Vocabulary And Study Skill

Use these words and study moves before you practise Number system. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Number system
- Number
- system
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Number system without a clear example, method, or evidence.

Number system: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Number system.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Number system: Practice And Correction

Practice for Number system:

Main outcome: compute numbers in different base systems and to do arithmetic operations based on binary numbers.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Number system: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Number system.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Number system: Outcome Check

Outcome check for Number system: Compute numbers in different base systems and to do arithmetic operations based on binary numbers.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Boolean algebra and logic gates: Lesson Opener

Learning focus: Boolean algebra and logic gates

Country link: a safe Rwanda observation, model, data table, or investigation for Boolean algebra and logic gates.

By the end of this topic, you should be able to:

- ☐ Identify different logic gates, theorems of Boolean algebra and evaluate Boolean expressions ☐
- Utilise laws of Boolean algebra on Boolean expressions and draw a simple logic circuit using logic gates

Inquiry questions:

- How can boolean algebra and logic gates help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Boolean algebra and logic gates: Learn

Boolean algebra and logic gates is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Boolean algebra and logic gates
- Boolean
- algebra
- logic
- gates
- concept
- observation
- evidence

Boolean algebra and logic gates: Worked Example

Investigation model for Boolean algebra and logic gates: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Boolean algebra and logic gates: Vocabulary And Study Skill

Use these words and study moves before you practise Boolean algebra and logic gates. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Boolean algebra and logic gates
- Boolean
- algebra
- logic
- gates
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Boolean algebra and logic gates without a clear example, method, or evidence.

Boolean algebra and logic gates: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Boolean algebra and logic gates.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Boolean algebra and logic gates: Practice And Correction

Practice for Boolean algebra and logic gates:

Main outcome: ☐ Identify different logic gates, theorems of Boolean algebra and evaluate Boolean expressions ☐ Utilise laws of Boolean algebra on Boolean expressions and draw a simple logic circuit using logic gates.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Boolean algebra and logic gates: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Boolean algebra and logic gates.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Boolean algebra and logic gates: Outcome Check

Outcome check for Boolean algebra and logic gates: ☐ Identify different logic gates, theorems of Boolean algebra and evaluate Boolean expressions ☐ Utilise laws of Boolean algebra on Boolean expressions and draw a simple logic circuit using logic gates.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control statements and one dimension array: Lesson Opener

Learning focus: Control statements and one dimension array

Country link: a safe Rwanda observation, model, data table, or investigation for Control statements and one dimension array.

By the end of this topic, you should be able to:

- derive logic in algorithm which include Control Statements and to handle one dimension array in algorithm

Inquiry questions:

- How can control statements and one dimension array help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control statements and one dimension array: Learn

Control statements and one dimension array is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Control statements and one dimension array
- Control
- statements
- dimension
- array
- concept
- observation
- evidence

Control statements and one dimension array: Worked Example

Worked example for Control statements and one dimension array: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Control statements and one dimension array: Vocabulary And Study Skill

Use these words and study moves before you practise Control statements and one dimension array. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Control statements and one dimension array
- Control
- statements
- dimension
- array
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Control statements and one dimension array without a clear example, method, or evidence.

Control statements and one dimension array: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Control statements and one dimension array.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Control statements and one dimension array: Practice And Correction

Practice for Control statements and one dimension array:

Main outcome: derive logic in algorithm which include Control Statements and to handle one dimension array in algorithm.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Control statements and one dimension array: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Control statements and one dimension array.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control statements and one dimension array: Outcome Check

Outcome check for Control statements and one dimension array: Derive logic in algorithm which include Control Statements and to handle one dimension array in algorithm.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Expression and Operator in C++ language: Lesson Opener

Learning focus: Expression and Operator in C++ language

Country link: a safe Rwanda observation, model, data table, or investigation for Expression and Operator in C++ language.

By the end of this topic, you should be able to:

- apply expressions and operators in C++ language

Inquiry questions:

- How can expression and operator in c++ language help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Expression and Operator in C++ language: Learn

Expression and Operator in C++ language is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Expression and Operator in C++ language
- Expression
- Operator
- language
- concept
- observation
- evidence
- safety

Expression and Operator in C++ language: Worked Example

Worked example for Expression and Operator in C++ language: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Expression and Operator in C++ language: Vocabulary And Study Skill

Use these words and study moves before you practise Expression and Operator in C++ language. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Expression and Operator in C++ language
- Expression
- Operator
- language
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Expression and Operator in C++ language without a clear example, method, or evidence.

Expression and Operator in C++ language: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Expression and Operator in C++ language.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Expression and Operator in C++ language: Practice And Correction

Practice for Expression and Operator in C++ language:

Main outcome: apply expressions and operators in C++ language.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Expression and Operator in C++ language: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Expression and Operator in C++ language.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Expression and Operator in C++ language: Outcome Check

Outcome check for Expression and Operator in C++ language: Apply expressions and operators in C++ language.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control Statements in C++: Lesson Opener

Learning focus: Control Statements in C++

Country link: a safe Rwanda observation, model, data table, or investigation for Control Statements in C++.

By the end of this topic, you should be able to:

- use control statements in C++ program to implement branching and iterations

Inquiry questions:

- How can control statements in c++ help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control Statements in C++: Learn

Control Statements in C++ is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Control Statements in C++
- Control
- Statements
- concept
- observation
- evidence
- safety

Control Statements in C++: Worked Example

Worked example for Control Statements in C++: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Control Statements in C++: Vocabulary And Study Skill

Use these words and study moves before you practise Control Statements in C++. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Control Statements in C++
- Control
- Statements
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Control Statements in C++ without a clear example, method, or evidence.

Control Statements in C++: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Control Statements in C++.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Control Statements in C++: Practice And Correction

Practice for Control Statements in C++:

Main outcome: use control statements in C++ program to implement branching and iterations.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Control Statements in C++: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Control Statements in C++.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Control Statements in C++: Outcome Check

Outcome check for Control Statements in C++: Use control statements in C++ program to implement branching and iterations.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Function in C++ language: Lesson Opener

Learning focus: Function in C++ language

Country link: a safe Rwanda observation, model, data table, or investigation for Function in C++ language.

By the end of this topic, you should be able to:

- define and use functions in C++ program

Inquiry questions:

- How can function in c++ language help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Function in C++ language: Learn

Function in C++ language is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Function in C++ language
- Function
- language
- concept
- observation

Function in C++ language: Worked Example

Worked example for Function in C++ language: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Function in C++ language: Vocabulary And Study Skill

Use these words and study moves before you practise Function in C++ language. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Function in C++ language
- Function
- language
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Function in C++ language without a clear example, method, or evidence.

Function in C++ language: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Function in C++ language.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Function in C++ language: Practice And Correction

Practice for Function in C++ language:

Main outcome: define and use functions in C++ program.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Function in C++ language: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Function in C++ language.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Function in C++ language: Outcome Check

Outcome check for Function in C++ language: Define and use functions in C++ program.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays in C++: Lesson Opener

Learning focus: Arrays in C++

Country link: a safe Rwanda observation, model, data table, or investigation for Arrays in C++.

By the end of this topic, you should be able to:

- use arrays and strings in a C++ program

Inquiry questions:

- How can arrays in c++ help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays in C++: Learn

Arrays in C++ is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Arrays in C++
- Arrays
- concept
- observation
- evidence
- safety
- conclusion

Arrays in C++: Worked Example

Worked example for Arrays in C++: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Arrays in C++: Vocabulary And Study Skill

Use these words and study moves before you practise Arrays in C++. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Arrays in C++
- Arrays
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Arrays in C++ without a clear example, method, or evidence.

Arrays in C++: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Arrays in C++.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Arrays in C++: Practice And Correction

Practice for Arrays in C++:

Main outcome: use arrays and strings in a C++ program.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Arrays in C++: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Arrays in C++.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Arrays in C++: Outcome Check

Outcome check for Arrays in C++: Use arrays and strings in a C++ program.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

HTML: Lesson Opener

Learning focus: HTML

Country link: a safe Rwanda observation, model, data table, or investigation for HTML.

By the end of this topic, you should be able to:

- build standards compliant web pages using HTML,

Inquiry questions:

- How can html help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

HTML: Learn

HTML is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- HTML
- concept
- observation
- evidence
- safety
- conclusion

HTML: Worked Example

Worked example for HTML: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

HTML: Vocabulary And Study Skill

Use these words and study moves before you practise HTML. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- HTML
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about HTML without a clear example, method, or evidence.

HTML: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for HTML.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

HTML: Practice And Correction

Practice for HTML:

Main outcome: build standards compliant web pages using HTML,.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

HTML: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from HTML.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

HTML: Outcome Check

Outcome check for HTML: Build standards compliant web pages using HTML,.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cascading style sheet: Lesson Opener

Learning focus: Cascading style sheet

Country link: a safe Rwanda observation, model, data table, or investigation for Cascading style sheet.

By the end of this topic, you should be able to:

- build standards compliant web pages using CSS

Inquiry questions:

- How can cascading style sheet help you solve a real problem in Computer Science?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cascading style sheet: Learn

Cascading style sheet is learned by connecting a computing concept to a clear problem, exact steps, and a checked output.

Core idea:

- Name the computing object first: data, algorithm, program, web page, database, network, device, or security control.
- Break the task into steps before using a computer.
- Use correct technical vocabulary such as input, process, output, variable, file, table, tag, command, or protocol where it fits.
- Test the result with a small example and correct one error.
- Save or present the work in a way another learner can inspect.

Key words:

- Cascading style sheet
- Cascading
- style
- sheet
- concept
- observation
- evidence
- safety

Cascading style sheet: Worked Example

Worked example for Cascading style sheet: Build a small checked computing output.

1. Define the task in one sentence.
2. List the inputs needed, such as text, numbers, records, tags, commands, or devices.
3. Write the steps before using the computer.
4. Create a small version of the output: a web page, algorithm trace, table, query, program fragment, network diagram, or security checklist.
5. Test with one example and correct one error.

Answer check: the output should match the task, use correct vocabulary, and be easy for another learner to inspect.

Cascading style sheet: Vocabulary And Study Skill

Use these words and study moves before you practise Cascading style sheet. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Cascading style sheet
- Cascading
- style
- sheet
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Cascading style sheet without a clear example, method, or evidence.

Cascading style sheet: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwanda observation, model, data table, or investigation for Cascading style sheet.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Cascading style sheet: Practice And Correction

Practice for Cascading style sheet:

Main outcome: build standards compliant web pages using CSS.

1. Write the purpose of the computing task in one sentence.
2. Create a small algorithm, code fragment, table, diagram, web page structure, database record, or network/security checklist.
3. Test it with one example and write one correction.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Cascading style sheet: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Cascading style sheet.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Cascading style sheet: Outcome Check

Outcome check for Cascading style sheet: Build standards compliant web pages using CSS.

1. State the computing task or problem.
2. Show the algorithm, code fragment, table, web page structure, database object, diagram, or security control.
3. Test with one example and write one correction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Computer fundamental: Review Clinic 1

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Computer fundamental that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Safe and ethical use of computer: Review Clinic 2

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Safe and ethical use of computer that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Number system: Review Clinic 3

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Number system that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Boolean algebra and logic gates: Review Clinic 4

Use this review clinic to strengthen Computer Science without copying earlier answers.

1. Choose one idea from Boolean algebra and logic gates that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Computer Science.

- Computer fundamental: Explain this term using an example from the book, your school, or your community.
- Safe and ethical use of computer: Explain this term using an example from the book, your school, or your community.
- Number system: Explain this term using an example from the book, your school, or your community.
- Boolean algebra and logic gates: Explain this term using an example from the book, your school, or your community.
- Control statements and one dimension array: Explain this term using an example from the book, your school, or your community.
- Expression and Operator in C++ language: Explain this term using an example from the book, your school, or your community.
- Control Statements in C++: Explain this term using an example from the book, your school, or your community.
- Function in C++ language: Explain this term using an example from the book, your school, or your community.
- Arrays in C++: Explain this term using an example from the book, your school, or your community.
- HTML: Explain this term using an example from the book, your school, or your community.
- Cascading style sheet: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Computer Science answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Computer Science that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.